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PROVISION OF A MEDICAL RISK ASSESSMENT FOR NONCLASSICAL 21-HYDROXYLASE DEFICIENCY

Abstract

The disclosure relates to an automated system for processing physiological parameter measurements for providing a medical risk assessment, wherein it comprises: a module for receiving physiological parameter measurements (p1 . . . p15) for a female to be assessed; and a module for calculating a nonclassical 21-hydroxylase deficiency score (SNC21OHD) for the female to be assessed, according to a model which gives the score (SNC21OHD) on the basis of the measurements (p1 . . . p15) received. The physiological parameters comprise at least the following steroids: 21-deoxycortisol; 21-deoxycorticosterone; 11 β -hydroxyandrostenedione; 17-OH-progesterone; androstenedione; corticosterone; and testosterone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This present application is a national stage application of International Patent Application No. PCT/FR2022/050759, filed Apr. 21, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure falls within the field of Nonclassical 21-hydroxylase deficiency (NC21OHD). More particularly, the present disclosure relates to systems, methods and computer programs for providing an assessment of a medical risk of nonclassical 21-hydroxylase deficiency (NC21OHD), and a method for assessing a medical risk of nonclassical 21-hydroxylase deficiency (NC21OHD).

BACKGROUND

[0003] The aim of the article “Androgen excess and diagnostic steroid biomarkers for nonclassical 21-hydroxylase deficiency without cosyntropin stimulation” by TURCU et al. published in the European Journal of Endocrinology, 183 63-71, in 2020, was to identify steroids that would allow reliable diagnosis of nonclassical 21-hydroxylase deficiency (NC21OHD). To this end, different steroids were measured for each female in a cohort and, using logistic regression modelling, three steroids were identified as being particularly discriminating in combination: 21-deoxycortisol, 17-OH-progesterone and corticosterone.

[0004] The article stresses that the measurements must be taken at the beginning of the follicular phase of the menstrual cycle, which is difficult to implement in some cases.

[0005] In addition, the model based on the combination of the three steroids mentioned above has not been validated on an independent cohort, so the robustness of this model may be questionable.

SUMMARY

[0006] The aim of the disclosure is to overcome some or all of these problems.

[0007] To this end, an automated system is proposed for processing physiological parameters measurements for providing a medical risk assessment, wherein it comprises: [0008] a module for receiving physiological parameters measurements for a female to be assessed; and [0009] a module for calculating a nonclassical 21-hydroxylase deficiency score for the female to be assessed, according to a model giving the score from the measurements received; the physiological parameters comprising at least the following steroids: [0010] 21-deoxycortisol, [0011] 21-deoxycorticosterone; [0012] 11 β -hydroxyandrostenedione; [0013] 17-OH-progesterone; [0014] androstenedione; [0015] corticosterone; and [0016] testosterone.

[0017] Surprisingly, the use of the above steroids allows a reliable and robust assessment of nonclassical 21-hydroxylase deficiency (NC21OHD), even when measurements are taken during the luteal phase of the menstrual cycle. In this way, the steroids selected mean that we are no longer dependent on the time at which the measurements are taken.

[0018] Optionally, the physiological parameters also comprise at least one of the following

steroids: [0019] pregnenolone; [0020] dehydroepiandrosterone; [0021] cortisol; [0022] cortisone; [0023] 11-deoxycorticosterone; [0024] 17-OH-pregnenolone; [0025] aldosterone; and [0026] progesterone.

[0027] Optionally, the physiological parameters may also comprise steroids only.

[0028] Optionally, the model giving the nonclassical 21-hydroxylase deficiency score is previously trained by machine learning.

[0029] Optionally, the model giving the nonclassical 21-hydroxylase deficiency score comprises a linear combination of the measurements of the physiological parameters.

[0030] Optionally, at least one of the steroids was measured during a luteal phase of a menstrual cycle of the female to be assessed.

[0031] Optionally, the automated system also comprises: [0032] a module for comparing the nonclassical 21-hydroxylase deficiency score with a predefined threshold, for example 0.5; and [0033] a module for calculating, when the nonclassical 21-hydroxylase deficiency score is on a predefined side of the threshold, a polycystic ovary syndrome score of the female to be assessed, according to a model giving the polycystic ovary syndrome score from at least some of the measurements received.

[0034] Optionally, the polycystic ovary syndrome score may also be obtained from at least the steroid measurements: 11 β -hydroxyandrostenedione and androstenedione, preferably from measurements of the steroids: testosterone and corticosterone.

[0035] A method for obtaining a model for an automated system according to the disclosure is also proposed, comprising: [0036] for each of several female in a cohort, obtaining measurements of the physiological parameters of a female and a status indicating that this female either has the nonclassical 21-hydroxylase deficiency or does not have this deficiency; and [0037] determining the model, by training the model using a training algorithm on the basis of measurements of the physiological parameters of the female in the cohort.

[0038] Optionally, all the steroids were measured during a follicular phase of a menstrual cycle.

[0039] Also proposed is a method for automated processing of physiological parameters measurements in order to provide an assessment of a medical risk, wherein it comprises: [0040] receiving measurements of physiological parameters of a female to be assessed; and [0041] using a model to provide a nonclassical 21-hydroxylase deficiency score for the female to be assessed, based on the measurements received;

the physiological parameters comprising at least the following steroids: [0042] 21-deoxycortisol; [0043] 21-deoxycorticosterone; [0044] 11 β -hydroxyandrostenedione; [0045] 17-OH-progesterone; [0046] androstenedione; [0047] corticosterone; and [0048] testosterone.

[0049] Optionally, the physiological parameters further comprise at least one of the following steroids: [0050] pregnenolone; [0051] dehydroepiandrosterone; [0052] cortisol; [0053] cortisone; [0054] 11-deoxycorticosterone; [0055] 17-OH-pregnenolone; [0056] aldosterone; and [0057] progesterone.

[0058] Optionally, the physiological parameters comprise steroids only.

[0059] Optionally, the model giving the nonclassical 21-hydroxylase deficiency score is previously trained by machine learning.

[0060] Optionally, the model giving the nonclassical 21-hydroxylase deficiency score comprises a linear combination of the measurements of the physiological parameters.

[0061] Optionally, at least one of the steroids was measured during a luteal phase of a menstrual cycle of the female to be assessed.

[0062] Optionally, the method also comprises: [0063] comparing the nonclassical 21-hydroxylase deficiency score with a predefined threshold, for example 0.5; and [0064] calculating, when the nonclassical 21-hydroxylase deficiency score is on a predefined side of the threshold, a polycystic ovary syndrome score of the female to be assessed, according to a model giving the score of polycystic ovary syndrome from at least one part of the measurements received.

[0065] Optionally, the polycystic ovary syndrome score may also be obtained from at least the measurements of the steroid: 11β -hydroxyandrostenedione and androstenedione, preferably also from measurements of the steroids: testosterone and corticosterone.

[0066] Also proposed is a computer program that may be downloaded from a communication network and/or recorded on a computer-readable medium, wherein it comprises instructions for executing a method according to the disclosure, when the computer program is executed on a computer.

[0067] Also proposed is a method for assessing a medical risk of a nonclassical 21-hydroxylase deficiency in a female to be assessed, comprising: [0068] collecting physiological parameters from the female to be assessed, the physiological parameters comprising at least the following steroids: 21-deoxycortisol, 21-deoxycorticosterone, 11β -hydroxyandrostenedione, 17-OH-progesterone, androstenedione, corticosterone and testosterone; [0069] calculating a nonclassical 21-hydroxylase deficiency score by introducing these physiological parameters into an automated system according to the disclosure; and [0070] comparing this score with a threshold, the female to be assessed being considered to be at risk of nonclassical 21-hydroxylase deficiency if the score is below the threshold.

[0071] Optionally, at least one of the steroids is measured during a luteal phase of a menstrual cycle of the female to be assessed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0072] The disclosure will be better understood with the aid of the following description, given only by way of example and made with reference to the attached drawings wherein:

[0073] FIG. 1 is a simplified view of an automated data processing system according to the disclosure for providing an assessment of a medical risk, in particular of nonclassical 21-hydroxylase deficiency (NC21OHD), on the basis of physiological parameters of a female to be assessed,

[0074] FIG. 2 is a block diagram illustrating the steps of a method according to the disclosure for processing data to provide an assessment of a medical risk, in particular of nonclassical 21-hydroxylase deficiency (NC21OHD),

[0075] FIG. 3 groups together several views of the system shown in FIG. 1, displaying information relating to the assessment carried out, in several different scenarios,

[0076] FIG. 4 is a block diagram illustrating the steps in a method for obtaining an M.sub.NC21OHD model for assessing a medical risk of nonclassical 21-hydroxylase deficiency (NC21OHD) and an M.sub.PCOS model for assessing a medical risk of polycystic ovary syndrome (PCOS),

[0077] FIG. 5 is a score plot resulting from an Orthogonal Partial Least Squares-Discriminant Analysis (OPLS-DA) carried out on a main development cohort to obtain the M.sub.NC21OHD model of the system shown in FIG. 1,

[0078] FIG. 6 is a score plot after using the M.sub.NC21OHD model on a validation cohort,

[0079] FIG. 7 repeats the score plot from FIG. 6, without the measurements from the validation cohort of female with a polycystic ovary syndrome (PCOS), and distinguishing between the measurements from control female and the measurements from female with nonclassical 21-hydroxylase deficiency (NC21OHD),

[0080] FIG. 8 repeats the score plot from FIG. 7, with the measurements from the validation cohort of female with polycystic ovary syndrome (PCOS),

[0081] FIG. 9 repeats the score plot from FIG. 6, isolating the female for whom at least one physiological parameter was measured in the luteal phase of their menstrual cycle,

[0082] FIG. **10** is a Variable Importance in Projection (VI P) plot of steroids used in the M.sub.NC2IOHD model,

[0083] FIG. **11** is a Variable Importance in Projection plot of the steroids used by the M.sub.PCOS model, and

[0084] FIG. **12** is a score plot resulting from a Partial Least Squares Discriminant Analysis (OPLS-DA) carried out on a development cohort referred to as annex, to obtain the M.sub.PCOS model.

DETAILED DESCRIPTION

[0085] With reference to FIG. **1**, an example of an automated data processing system **100** according to the disclosure will now be described. More specifically, as will become apparent later, the system **100** is designed to provide an assessment of a medical risk that a female has nonclassical 21-hydroxylase deficiency (NC21OHD). This assessment is provided, for example, in the form of a S.sub.NC2IOHD score for nonclassical 21-hydroxylase deficiency (NC21OHD). In this way, the system **100** provides clinical decision support, for example for the practitioner.

[0086] The system **100** firstly comprises a man/machine interface **102** comprising at least one information input device and at least one information presentation device, for example a visual presentation device such as a screen. For example, the man/machine interface **102** comprises a touch screen acting as both an information input device and an information presentation device. For example, the system **100** is a tablet computer or a smartphone.

[0087] The system **100** also comprises a data reception module **104**, designed to receive physiological parameters measurements of the female to be assessed, the physiological parameters comprise at least the following steroids: 21-deoxycortisol (21 DF), 21-deoxycorticosterone (21DB), 11 β -hydroxyandrostenedione (11bOHA), 17-OH-progesterone (17OHP), androstenedione (A), corticosterone (B), and testosterone (T). Preferably, the physiological parameters also comprise at least one of the following steroids: pregnenolone (Preg), dehydroepiandrosterone (DHEA), cortisol (F), cortisone (E), 11-deoxycorticosterone (DOC), 17-OH-pregnenolone (17-OHPreg), aldosterone (ALDO), and progesterone (P). Even more preferably, the physiological parameters comprise only steroids.

[0088] In the example shown in FIG. **1**, the measurements of the fifteen steroids listed above are received by the data reception module **104**, and noted p1 . . . p15.

[0089] For example, the data reception module **104** is designed to cooperate with the man/machine interface **102** to receive the measurements p1 . . . p15. For example, a user of the system **100** may enter the measurements p1 . . . p15 by hand via the man/machine interface **102**. Alternatively, all or some of the measurements p1 . . . p15 may be received from a remote device (not shown), such as a computer, via a wireless or wired connection.

[0090] The physiological parameters form a multidimensional space (of dimension fifteen when all the steroids indicated above are used) of which each set of measurements p1 . . . p15 is a point.

[0091] The system **100** also comprises a module **106** for calculating the S.sub.NC2IOHD score of the female to be assessed. This calculation module **106** is designed to calculate the S.sub.NC2IOHD score according to a M.sub.NC2IOHD model, for example previously trained by machine learning.

[0092] For example, the M.sub.NC2IOHD model is a linear combination, so the S.sub.NC2IOHD score is a linear combination of the received measurements p1 . . . p15. In this linear combination, the measurements p1 . . . p15 are assigned with weights w1 . . . w15 respectively, and the weighted measurements p1 . . . p15 are summed to obtain the S.sub.NC2IOHD score. In this case, the S.sub.NC2IOHD score is given by the following formula:

$$[00001] S_{\text{NC21OHD}} = \sum_{n=1}^N w_n \cdot p_n + [\text{Math} \cdot 1]$$

where N is the number of physiological parameters measured (for example fifteen in the example in FIG. **1**), and s is a constant representing a residual. This constant may be zero.

[0093] As will be explained in more detail later, the machine learning comprises determining the

weights $w_1 \dots w_{15}$ allowing the S.sub.NC2IOHD score to be representative of the risk of nonclassical 21-hydroxylase deficiency (NC2IOHD) in the female to be assessed.

[0094] The machine learning may be carried out outside the automated system **100**. In this case, the data reception module **104** may also be designed to receive an update containing the weights $w_1 \dots w_{15}$. This update is received, for example, from a remote device (not shown), such as a computer, via a wireless or wired connection.

[0095] Alternatively, the M.sub.NC2IOHD model could comprise a neural network.

[0096] Generally speaking, the M.sub.NC2IOHD model implements a function including parameters and the prior machine learning of the M.sub.NC2IOHD model comprises determining these parameters, allowing the S.sub.NC2IOHD score to be representative of the risk of nonclassical 21-hydroxylase deficiency (NC2IOHD).

[0097] Preferably, the M.sub.NC2IOHD model is pre-trained so that the S.sub.NC2IOHD score lies within a predefined range, for example between 0 and 1.

[0098] In addition, the system **100** comprises, for example, a module **108** for comparing the S.sub.NC2IOHD score with a predefined threshold T.sub.NC2IOHD, for example 0.5. When the S.sub.NC2IOHD score is on a predefined side (above or well below) the T.sub.NC2IOHD threshold, the female to be assessed is considered to be at high risk of nonclassical 21-hydroxylase deficiency (NC2IOHD), and at low risk on the other side. The result of this comparison provides a D.sub.NC2IOHD assessment of the risk of nonclassical deformed 21-hydroxylase deficiency (NC2IOHD). For example, the D.sub.NC2IOHD risk assessment is binary and indicates either a high risk of nonclassical deformed 21-hydroxylase deficiency (NC2IOHD) when the S.sub.NC2IOHD score is above the T.sub.NC2IOHD threshold, or a low risk of nonclassical deformed 21-hydroxylase deficiency (NC2IOHD) when the S.sub.NC2IOHD score is below the T.sub.NC2IOHD threshold, or vice versa. Alternatively, the D.sub.NC2IOHD risk assessment could be more gradual, for example additionally indicating a very low risk when the S.sub.NC2IOHD score is well below the T.sub.NC2IOHD threshold (e.g. below 0.25) and/or a very high risk when the S.sub.NC2IOHD score is well above the T.sub.NC2IOHD threshold (e.g. above 0.75).

[0099] The system **100** also comprises a module **110** for calculating, when the S.sub.NC2IOHD score is on the predefined side of the T.sub.NC2IOHD threshold indicating a low risk of nonclassical 21-hydroxylase deficiency (NC2IOHD), a S.sub.pcos polycystic ovary syndrome (PCOS) score for the female to be assessed, on the basis of at least some of the measurements $p_1 \dots p_{15}$, for example all of them, or even just all of these measurements. This calculation module **110** is designed to calculate the S.sub.pcos score according to an M.sub.PCOS model, for example previously trained by machine learning.

[0100] For example, the M.sub.PCOS model is a linear combination, so the S.sub.pcos score is a linear combination of the $p_1 \dots p_{15}$ measurements used. In this linear combination, the measurements $p_1 \dots p_{15}$ used are respectively assigned weights $w'_1 \dots w'_{15}$, and the weighted measurements $p_1 \dots p_{15}$ are summed to obtain the S.sub.pcos score. In this case, the S.sub.NC2IOHD score is given by the following formula:

$$[00002] S_{PCOS} = \text{Math.}_{n=1}^N w'_n \text{Math. } p_n + s' \quad [\text{Math. } 2]$$

where N is the number of measured physiological parameters used (e.g. fifteen in the example in FIG. 1), and s' is a constant representing a residual. This constant may be zero.

[0101] As will be explained in more detail later, the machine learning comprises determining the weights $w'_1 \dots w'_{15}$ allowing the S.sub.PCOS score to be representative of the risk that the female being assessed has polycystic ovary syndrome (PCOS).

[0102] The machine learning may be carried out outside the automated system **100**. In this case, the data reception module **104** may also be designed to receive an update containing the weights $w'_1 \dots w'_{15}$. This update is received, for example, from a remote device (not shown), such as a computer,

via a wireless or wired connection.

[0103] Alternatively, the M.sub.PCOS model could comprise a neural network.

[0104] In general, the M.sub.PCOS model implements a function including the parameters and the prior machine learning of the M.sub.PCOS model comprises the determination of these parameters allowing the S.sub.PCOS score to be representative of the risk of developing polycystic ovary syndrome (PCOS).

[0105] Preferably, the M.sub.PCOS model is pre-trained so that the S.sub.PCOS score lies within a predefined range, for example between 0 and 1.

[0106] In addition, the system **100** comprises, for example, a module **112** for comparing the S.sub.PCOS score with a predefined T.sub.PCOS threshold, for example 0.5. When the S.sub.PCOS score is on a predefined side (above or well below) the T.sub.PCOS threshold, the female to be assessed is considered to be at high risk of polycystic ovary syndrome, and at low risk on the other side. The result of this comparison provides a D.sub.PCOS assessment of the risk of polycystic ovary syndrome. For example, the D.sub.PCOS risk assessment is binary and indicates either a low risk of polycystic ovary syndrome (PCOS) when the S.sub.PCOS score is above the T.sub.PCOS threshold, or a high risk of polycystic ovary syndrome (PCOS) when the S.sub.PCOS score is below the T.sub.PCOS threshold, or vice versa. Alternatively, the D.sub.PCOS risk assessment could be more gradual, for example by also indicating a very low risk when the S.sub.PCOS score is well below the T.sub.PCOS threshold (e.g. below 0.25) and/or a very high risk when the S.sub.PCOS score is well above the T.sub.PCOS threshold (e.g. above 0.75).

[0107] The system **100** also comprises an information presentation module **116** designed to present the S.sub.NC2IOHD score and/or the D.sub.NC2IOHD risk assessment to a user of the system **100**. The presentation module **116** may also be designed to present the S.sub.pcos score and/or the D.sub.PCOS risk assessment.

[0108] The presentation module **116** is designed, for example, to cooperate with the man/machine interface **102**. For example, the presentation module **116** is designed to cause information to be displayed on the man/machine interface **102**, for example on the touch screen when such a screen is used.

[0109] Furthermore, the system **100** comprises, for example, a computer system comprising a processing unit **118** (such as one or more microprocessors) and a main memory **120** coupled to the processing unit **118**. A computer program **124** containing computer program instructions is configured to be loaded, either in its entirety or sequentially in chunks, into the main memory **120**. The loaded instructions may then be executed by the processing unit **118**. For example, the system **100** also comprises a memory of mass **122** wherein the computer program **124** is stored before being loaded into the main memory **120**.

[0110] For example, the modules described above are implemented in the computer program in the form of software modules.

[0111] For example, particularly where the system **100** is a smartphone, the computer program **124** may be in the form of an application (APP), for example available from the Apple (trademark) and/or Android (trademark) app shop.

[0112] Alternatively, all or some of the modules could be implemented in the form of hardware modules, i.e. in the form of an electronic circuit, for example micro-wired, not involving a computer program.

[0113] With reference to FIG. 2, an example **200** of an automated data processing method, in the context of the example system **100** of FIG. 1, will now be described.

[0114] During a step **202**, the reception module **104** receives the measurements $p_1 \dots p_{15}$.

[0115] In a step **204**, the calculation module **106** implementing the M.sub.NC2IOHD model calculates the S.sub.NC2IOHD score from the measurements $p_1 \dots p_{15}$ received.

[0116] In a step **206**, the comparison module **108** compares the S.sub.NC2IOHD score with the predefined threshold and determines the D.sub.NC2IOHD risk assessment.

[0117] In a step **208**, if the S.sub.NC2IOHD score is on the side of the T.sub.NC2IOHD threshold indicating a low risk of nonclassical 21-hydroxylase deficiency (NC21OHD), for example if the S.sub.NC2IOHD score is less than 0.5, the calculation module **110** implementing the M.sub.PCOS model calculates the Spcos score from at least some of the measurements p1 . . . p15 received.

[0118] In a step **210**, the comparison module **110** compares the S.sub.PCOS score with the T.sub.PCOS threshold and determines the D.sub.PCOS risk assessment.

[0119] During a step **212**, the information presentation module **116** presents the S.sub.NC2IOHD score and/or the D.sub.NC2IOHD risk assessment, as well as, if applicable, the S.sub.PCOS score and/or the D.sub.PCOS risk assessment.

[0120] With reference to FIG. 3, during step **212**, the presentation module **116** causes, for example, a sentence representing the D.sub.NC2IOHD risk assessment to be displayed, for example “high NC21OHD risk” or “low NC21OHD risk”, depending on the D.sub.NC2IOHD risk assessment. Still for example, the presentation module **116** causes the S.sub.NC2IOHD score and, preferably, the predefined threshold to be displayed. The latter is 0.5 in the example shown in FIG. 3. In addition, when the S.sub.NC2IOHD score is on the side of the T.sub.NC2IOHD threshold indicating a low risk of nonclassical 21-hydroxylase deficiency (NC21OHD), the presentation module **116** causes, for example, a sentence representing the D.sub.PCOS risk assessment of polycystic ovary syndrome (PCOS) to be displayed, for example “high PCOS risk” or “low PCOS risk”, depending on the D.sub.PCOS risk assessment. Still for example, the presentation module **116** causes the S.sub.PCOS score and, preferably, the T.sub.PCOS threshold to be displayed.

[0121] With reference to FIG. 4, a method **400** for obtaining the M.sub.NC2IOHD and M.sub.PCOS models for an automated system such as that shown in FIG. 1 will now be described.

[0122] In a step **402**, for each of several female in a so-called main development cohort, measurements p1 . . . p15 of physiological parameters of the woman are obtained, as well as a status of nonclassical 21-hydroxylase deficiency (NC21OHD) of this female. For example, for all the female in the main development cohort, measurements are taken during the follicular phase of the menstrual cycle. The physiological parameters comprise at least the following steroids: 21-deoxycortisol (21 DF), 21-deoxycorticosterone (21DB), 11 β -hydroxyandrostenedione (11bOHA), 17-OH-progesterone (17OHP), androstenedione (A), corticosterone (B), and testosterone (T). Preferably, the physiological parameters also comprise at least one of the following steroids: pregnenolone (Preg), dehydroepiandrosterone (DHEA), cortisol (F), cortisone (E), 11-deoxycorticosterone (DOC), 17-OH-pregnenolone (17-OHPreg), aldosterone (ALDO), and progesterone (P). Even more preferably, the physiological parameters comprise only steroids.

[0123] The status is binary and indicates either a nonclassical 21-hydroxylase deficiency (NC21OHD) or the absence of this deficiency. For the method **400** to function correctly, the status of at least one female indicates that this female has a nonclassical 21-hydroxylase deficiency (NC21OHD) and the status of at least one other female indicates the absence of this deficiency.

[0124] For some of the female in the main development cohort, the measurements for certain steroids may be missing.

[0125] In addition, in step **402**, measurements p1 . . . p15 of physiological parameters from each of several female in a validation cohort are obtained, together with a nonclassical 21-hydroxylase deficiency (NC21OHD) status for that female. Preferably, polycystic ovary syndrome (PCOS) status is also obtained for each female in the validation cohort.

[0126] In a step **404**, the measurements from the main development cohort are analyzed to look for statistically outlier measurements for one or more female in the main development cohort. In addition, each female whose measurements were statistically aberrant was removed from the main development cohort. To carry out this data analysis, for example, the measurements are first centered and reduced. Principal Component Analysis (PCA) may then be used, for example, to search for one or more sets of statistically outlier measurements.

[0127] In a step **406**, the M.sub.NC2IOHD model is built by machine learning from measurements

of physiological parameters for the female in the main development cohort. The aim of machine learning is for the M.sub.NC2IOHD model to provide a S.sub.NC2IOHD score that reliably predicts the status of female in the main development cohort and, subsequently, of a new female whose measurements of steroids are provided to the M.sub.NC2IOHD model. A training algorithm is generally used for this. This is designed to adjust parameters of the M.sub.NC2IOHD model from training data comprising input data and output data, so that the M.sub.NC2IOHD model provides, from known input data, the known output data.

[0128] For example, the M.sub.NC2IOHD model is a linear combination of measurements of physiological parameters and machine learning is designed to determine weights of this linear combination, so that the linear combination reliably predicts the statuses of female in the main development cohort.

[0129] For example, an Orthogonal Partial Least Squares-Discriminant Analysis (OPLS-DA) is used.

[0130] Taking steroid measurements on female is a difficult operation to implement, so measurements may generally only be obtained for a small number of female, for example a few dozen to a few hundred. In addition, some measurements may be missing. In addition, it is reasonable to assume that the steroids are a priori multi-collinear. The OPLS-DA analysis therefore seems particularly appropriate as it is effective in the case of multicollinearity, even in a small cohort. It is also robust, meaning that it may be carried out even with some measurements missing.

[0131] As is well known, by noting X a matrix grouping the measurements of physiological parameters for the female in the main development cohort and Y a vector grouping the status of these females, the OPLS-DA analysis decomposes the X and Y matrices as follows:

$$[00003] \quad X = TP^T + E_X \quad [\text{Math. 3}] \quad Y = UQ^T + E_Y$$

where T and U are the scores of the X and Y matrices respectively, P and Q are the loadings of the X and Y matrices respectively, and E_X and E_Y are the residuals, which the decomposition seeks to make as small as possible.

[0132] The weights of the linear combination, noted w_n with $n=1 \dots N$ where N is the number of physiological parameters, are then the components of a vector W defined as follows:

$$[00004] \quad T = XW^* \quad [\text{Math. 4}]$$

W^* being the trans-conjugate vector of the vector W.

[0133] The M.sub.NC2IOHD model is therefore the linear combination of the measurements of the physiological parameters, with the weights corresponding respectively to the components w_n of the vector W.

[0134] In a step **408**, the M.sub.NC2IOHD model may be evaluated in order to validate or invalidate it.

[0135] The evaluation may be an internal evaluation (i.e. based on the main development cohort) and/or an external evaluation (i.e. based on the validation cohort).

[0136] For example, the internal evaluation may use one or more of the following: the coefficient of determination R^2 , the permutation test, the prediction coefficient Q^2 and the average accuracy (i.e. the proportion of correct predictions made by the M.sub.NC2IOHD model).

[0137] For example, the external evaluation uses the receiver operating characteristic (ROC) of the validation cohort. For example, an Area Under the Curve (AUC) of the ROC curve may be compared with a predefined threshold. The average accuracy may also be used for the external evaluation, based on the validation cohort (validation accuracy).

[0138] In step **410**, the M.sub.PCOS model is built by machine learning from measurements of the physiological parameters $p_1 \dots p_{15}$ for female in a so-called annex development cohort. This annex development cohort is made up, for example, of control female from the validation cohort and female with polycystic ovary syndrome (PCOS). The aim of machine learning is for the M.sub.PCOS model to provide a S.sub.PCOS score that reliably predicts the status of female in the

annex development cohort and, subsequently, of a new female whose steroid measurements are provided to the M.sub.PCOS model. A training algorithm is generally used for this. The latter is designed to adjust parameters of the M.sub.PCOS model from training data comprising input data and output data, so that the M.sub.PCOS model provides, from known input data, the known output data.

[0139] For example, the M.sub.PCOS model is a linear combination of measurements of physiological parameters and machine learning is designed to determine weights of this linear combination, so that the linear combination reliably allows to predict the statuses of female in the annex development cohort.

[0140] For example, an Orthogonal Partial Least Squares-Discriminant Analysis (OPLS-DA) is used.

[0141] In a step **412**, the M.sub.PCOS model may be evaluated in order to validate or invalidate it. For example, a cross-validation based on the annex development cohort may be carried out at the same time as the machine learning.

[0142] An example of implementation of the method **400** may be used is described below.

Stage 402: Obtaining the Measurements

[0143] In the example described, the main development cohort comprises 38 females, including 19 with nonclassical 21-hydroxylase deficiency (NC21OHD) and 19 so-called control female, i.e. with neither this deficiency nor polycystic ovary syndrome (PCOS). For its part, the validation cohort comprises, in the example described, 355 females, including 17 with nonclassical 21-hydroxylase deficiency (NC21OHD), 266 females with polycystic ovary syndrome (PCOS) and 72 so-called control females, i.e. female with none of these diseases.

[0144] In addition, in the example described, the fifteen steroids detailed above were measured in each female in the main development cohort, during the follicular phase of their menstrual cycle. These same fifteen steroids were measured in every female in the validation cohort, during the luteal phase of their menstrual cycle for some of the female and during the follicular phase for the others.

Step 404: Statistical Verification of the Measurements

[0145] In the example described, no statistically outlier measurements were found in the measurements of the main development cohort.

Step 406: Obtaining the M.SUB.NC21OHD .Model

[0146] The main development cohort is used to determine the M.sub.NC21OHD model from the measurements of the fifteen steroids. In the example described, the OPLS-DA analysis is used, so that the M.sub.NC21OHD model obtained is a linear combination of the measurements of the fifteen steroids. FIG. 5 is a score plot of the measurements of the females in the main development cohort. As may be seen, the measurements of the females with nonclassical 21-hydroxylase deficiency (NC21OHD) (dark grey dots) and the measurements of the control females (light grey dots) form two distinct groups, to the left and right of the central vertical line respectively.

Step 408: Validation of the M.SUB.NC21OHD.7 Model

[0147] FIG. 6 is a score plot of the measurements of the females in the main development cohort of (dark grey dots and light grey dots, as in FIG. 5) and the measurements of the females in the validation cohort (black stars), with no distinction for the latter between control females, the females with nonclassical 21-hydroxylase deficiency (NC21OHD) and the females with polycystic ovary syndrome (PCOS).

[0148] FIG. 7 is similar to FIG. 6, without the females with polycystic ovary syndrome (PCOS) and distinguishing the control females (light grey stars) from the females with nonclassical 21-hydroxylase deficiency (NC21OHD) (dark grey stars).

[0149] FIG. 8 is similar to FIG. 7, without the control females and the females with nonclassical 21-hydroxylase deficiency (NC21OHD), but with the females with polycystic ovary syndrome (PCOS) (black stars).

[0150] As may be seen in FIG. 6, FIG. 7 and FIG. 8, the M.sub.NC2IOHD model obtained allow the females with nonclassical 21-hydroxylase deficiency (NC21OHD) to be distinguished not only from control females, but also from females with polycystic ovary syndrome (PCOS).

[0151] FIG. 9 is a score plot of the measurements of the control female and having the polycystic ovary syndrome (PCOS) in the validation cohort, for whom steroids were measured during the luteal phase of their menstrual cycle. As may be seen, despite the measurements in the luteal phase, they are well recognized as being at low risk of nonclassical 21-hydroxylase deficiency (NC21OHD).

[0152] Surprisingly, a reliable assessment was obtained even for the females whose measurements were taken during the luteal phase of their menstrual cycle. In this way, the steroids selected mean that we are no longer dependent on the time at which the measurements are taken.

[0153] FIG. 10 is a Variable Importance in Projection (VIP) plot of the fifteen steroids in the M.sub.NC2IOHD model.

[0154] As may be seen from this FIG., the seven most important steroids are: 21-deoxycortisol (21DF), 21-deoxycorticosterone (21DB), 11 β -hydroxyandrostenedione (11bOHA), 17-OH-progesterone (17-OHP), androstenedione (A), corticosterone (B), and testosterone (T). It was even found that using of these seven steroids, without the eight others listed above, gave a fairly reliable assessment, even with measurements taken in the luteal phase.

[0155] It will be appreciated that new females may be added to the main development cohort to refine the M.sub.NC2IOHD model. Thus, a new OPLS-DA analysis (as implemented for example in the steps of the method 400 detailed above) may be carried out with the development cohort thus enriched with new females, in order to obtain updated weights for the M.sub.NC2IOHD model. These updated weights may then be transmitted to the systems 100 for updating.

Step 410: Obtaining the M.SUB.PCOS .Model

[0156] The annex development cohort consists of 266 females with a polycystic ovary syndrome (PCOS) and 72 control females from the validation cohort. This annex development cohort is used to obtain the M.sub.PCOS model.

[0157] FIG. 11 is a Variable Importance in Projection (VIP) plot of the fifteen steroids in the M.sub.PCOS model.

[0158] As may be seen in this FIG., the two most important steroids are androstenedione (A) and 11 β -hydroxyandrostenedione (11bOHA), possibly supplemented by pregnenolone (Preg) and 21-deoxycorticosterone (21DB).

[0159] As may be seen in FIG. 12, the M.sub.PCOS model obtained allow to distinguish the females with polycystic ovary syndrome (PCOS) (circles) from control females (stars).

[0160] In conclusion, it is clear that a system such as the one described above provides a reliable assessment of the risk of nonclassical 21-hydroxylase deficiency (NC21OHD).

[0161] It should be noted that the disclosure is not limited to the embodiments described above. In fact, it will appear to the person skilled in the art that various modifications may be made to the above-described embodiments, in the light of the teaching just disclosed.

[0162] In particular, the M.sub.PCOS model could be used independently of the M.sub.NC2IOHD model to provide an assessment of a medical risk of polycystic ovary syndrome (PCOD), for example in cases where a medical risk of nonclassical 21-hydroxylase deficiency (NC21OHD) in the female to be assessed has been ruled out by a method other than the use of the M.sub.NC2IOHD model.

[0163] In the detailed presentation of the disclosure given above, the terms used should not be interpreted as limiting the disclosure to the embodiments set out in this description, but should be interpreted to comprise all equivalents the anticipation of which is within the grasp of the person skilled in the art by applying his general knowledge to the implementation of the teaching just disclosed.

Claims

1. An automated system for processing physiological parameters measurements for providing a medical risk assessment, wherein it comprises: a module for receiving physiological parameters measurements (p1 . . . p15) for a female to be assessed; and a module for calculating a nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) for the female to be assessed, according to a model giving the nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) from the measurements (p1 . . . p15) received; the physiological parameters comprising at least the following steroids: 21-deoxycortisol (21DF); 21-deoxycorticosterone (21DB); 11 β -hydroxyandrostenedione (11bOHA); 17-OH-progesterone (17OHP); androstenedione (A); corticosterone (B); and testosterone (T).
2. The automated system of claim 1, wherein the physiological parameters further comprise at least one of the following steroids: pregnenolone (Preg); dehydroepiandrosterone (DHEA); cortisol (F); cortisone (E); 11-deoxycorticosterone (DOC); 17-OH-pregnenolone (170HPreg); aldosterone (ALDO); and progesterone (P).
3. The automated system according to claim 1, wherein the physiological parameters comprise steroids only.
4. The automated system according to claim 1, wherein the model giving the nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) is previously trained by machine learning.
5. The automated system according to claim 4, wherein the model giving the nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) comprises a linear combination of the measurements of the physiological parameters.
6. The automated system according to claim 1, wherein at least one of the steroids was measured during a luteal phase of a menstrual cycle of the female to be assessed.
7. The automated system according to claim 1, further comprising: a module for comparing the nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) with a predefined threshold, for example 0.5; and a module for calculating, when the nonclassical 21-hydroxylase deficiency score (S.sub.NC2IOHD) is on a predefined side of the threshold, a polycystic ovary syndrome score (S.sub.PCOS) of the female to be assessed, according to a model giving the polycystic ovary syndrome score (S.sub.PCOS) from at least some of the measurements (p1 . . . p15) received.
8. The automated system according to claim 7, wherein the score (S.sub.PCOS) of polycystic ovary syndrome is given from at least the steroid measurements: 11 β -hydroxyandrostenedione (11bOHA) and androstenedione (A), preferably also from measurements of the steroids: testosterone (T) and corticosterone (B).
9. A method for obtaining a model for an automated system according to claim 1, comprising: for each of several female in a cohort, obtaining measurements of the physiological parameters of a female and a status indicating that the female either has the nonclassical 21-hydroxylase deficiency or does not have this deficiency; and determining the model, by training the model using a training algorithm, on the basis of measurements of the physiological parameters of the female in the cohort.
10. The method of claim 9, wherein all steroids were measured during a follicular phase of a menstrual cycle.
11. The method for automated processing of physiological parameters measurements in order to provide an assessment of a medical risk, comprising: receiving measurements (p1 . . . p15) of physiological parameters of a female to be assessed; and using a model to provide a nonclassical 21-hydroxylase deficiency score (S) for the female to be assessed, based on the measurements (p1 . . . p15) received; the physiological parameters comprising at least the following steroids: 21-deoxycortisol (21DF); 21-deoxycorticosterone (21DB); 11 β -hydroxyandrostenedione (11bOHA); 17-OH-progesterone (17OHP); androstenedione (A); corticosterone (B); and testosterone (T).

- 12.** The method according to claim 11, wherein the physiological parameters further comprise at least one of the following steroids: pregnenolone (Preg); dehydroepiandrosterone (DHEA); cortisol (F); cortisone (E); 11-deoxycorticosterone (DOC); 17-OH-pregnenolone (17OHPreg); aldosterone (ALDO); and progesterone (P).
- 13.** The method according to claim 11, wherein the physiological parameters comprise steroids only.
- 14.** The method according to claim 11, wherein the model giving the nonclassical 21-hydroxylase deficiency score (S.sub.NC21OHD) is previously trained by machine learning.
- 15.** The method according to claim 11, wherein the model giving the nonclassical 21-hydroxylase deficiency score comprises a linear combination of the measurements of the physiological parameters.
- 16.** The method according to claim 11, wherein at least one of the steroids was measured during a luteal phase of a menstrual cycle of the female to be assessed.
- 17.** The method according to claim 11, further comprising: comprising the nonclassical 21-hydroxylase deficiency score (S.sub.NC21OHD) with a predefined threshold, for example 0.5; and calculating, when the nonclassical 21-hydroxylase deficiency score (S.sub.NC21OHD) is on a predefined side of the threshold, a score (S.sub.PCOS) of polycystic ovary syndrome of the female to be assessed, according to a model giving the score (S.sub.PCOS) of polycystic ovary syndrome from at least one part of the measurements (p1 . . . p15) received.
- 18.** The method according to claim 17, wherein the polycystic ovary syndrome score (S.sub.PCOS) is given from at least the measurements of the steroids: 11 β -hydroxyandrostenedione (11bOHA) and androstenedione (A), preferably from measurements of the steroids: testosterone (T) and corticosterone (B).
- 19.** A computer program downloadable from a communication network and/or recorded on a computer-readable medium, comprising instructions for executing a method according to claim 11, when the computer program is executed on a computer.
- 20.** A method for assessing a medical risk of a nonclassical 21-hydroxylase deficiency in a female to be assessed, comprising: collecting physiological parameters from the female to be assessed, the physiological parameters comprising at least the following steroids: 21-deoxycortisol (21DF), 21-deoxycorticosterone (21DB), 11 β -hydroxyandrostenedione (11bOHA), 17-OH-progesterone (17OHP), androstenedione (A), corticosterone (B), and testosterone (T); calculating a nonclassical 21-hydroxylase deficiency score (S) by introducing these physiological parameters into an automated system according to any one of claims 1 to 6; and comparing this score (S) with a threshold, the female to be assessed being considered to be at risk of nonclassical 21-hydroxylase deficiency if the score is below the threshold.
- 21.** The method according to claim 20, wherein at least one of the steroids is measured during a luteal phase of a menstrual cycle of the female to be assessed.
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